Research Assignment: Introduction to Data Science and Machine Learning

1: Data Science is the field of study that involves extracting knowledge and insights from noisy data to actions. while Machine Learning learn from data to create insights that improves performance to inform predictions.

the relationship between them:

Although Data Science and Machine Learning are related but they are different.

Data Science encompasses the entire life cycle of data management , analysis and structure big data. while Machine learning is a part of data science that provides predictive tools.

real life example:

Data Science	Machine Learning
1:Transportation: optimizes shipping road routes in real time	predicts delivery arrival times and forecast how storms to holiday rushes — will impact traffic jams.
2:Sports: accurately evaluates athletes performance	predicts the risks of future player injures and simulate match outcomes against specific opponents.
3:Gaming: improves online gaming experiences	Automatically adjusts game difficulty based on user skill level.

2:Data Science life cycle is systematic approach to extracting value from data.

1. **Problem Definition**: The first step is clearly defining the problem that needs to be solved.
2. **Data Collection**: involves systematic collection of data necessary to solve the defined problem.
3. **Data Cleaning:** Raw data is often messy and unstructured or incorrect format.
4. **Exploratory Data Analysis (EDA): To** find patterns and characteristics hidden in the data to use statistically and visual tools to explore patterns in data.
5. **Model Training:** With the selected model the machine learning lifecycle moves to model training process.
6. **Model Evaluation and Tuning:** involves rigorous testing against validation or test datasets to test accuracy of model on new unseen data.

7. **Model Deployment:** Now model is ready for deployment for real-world application. It involves integrating the predictive model with existing systems allowing business to use this for informed decision-making.
8. **Model Monitoring and Maintenance:** **Track model performance over time.**

which stage does machine learning fit and why?

stage 5 and 6: model training and model evaluation

why?

this is where the actual machine learning happens. instead of writing "if/else" code rules.

ML models are highly overfitting

3:Supervised learning is a type of machine learning where the algorithm is trained on a labeled dataset. While Unsupervised learning is a type of machine learning where the algorithm is provided with input data without explicit instructions on what to do with it.

Example of each:

Supervised example:

Stock Price Prediction: Forecasting the future prices of stocks by analyzing historical stock market data.

Unsupervised example:

Customer segmentation: This involves the identification of distinct groups and patterns within market segmentation, allowing companies to better understand their customer base.

4:Overfitting is a common challenge in machine learning that occurs when a model learns the training data too well, including its noise and irrelevant details.

Causes:

- **Insufficient training data**
- **Noisy data** with random variations.

Preventing overfitting:

- **Dropout (for neural networks)**
- **Data augmentation**
- **Early stopping**

5: Training data is the dataset used to teach a machine learning model. while Test data Once the model has learned from training data, we need new, unseen data to check if it has learned correctly.

this process splitting your dataset into training data and test data is absolutely necessary for several critical reasons.

6:**Healthcare: Predictive Analytics for Better Outcomes**

Data science has changed the way patient care is delivered with great impact. For example, Mayo Clinic and IBM Watson Health leverage predictive analytics in the analysis of patient records and clinical data.

This machine learning can be utilized in identifying potential health problems and in developing personal treatment programs, and plays a crucial role in making effective use of resources in a hospital. With the help of predictive insights, clinicians can act at an earlier stage, leading to better patient outcomes and cost savings.

Result:

- Earlier and more accurate diagnoses
- Faster clinical decision-making
- Personalized treatment strategies
- Improved patient outcomes

These data science real life case studies show how data analytics has its roots in efficiency, yet it creates significant, life-saving impacts.

Lifecycle parts covered:

- data collection analyzing complex large volumes of historical patient.
- model training implementing machine learning to identify potentail health risks and patterns.
- deployment provides real-time predictive directly to clinicians at the hospital to speedup decision-making